

Article

Flow Resistance and Heat Transfer During Flow Boiling of HFE-649 in an Annular Minigap [†]

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Abstract

This paper investigates flow resistance and heat transfer during flow boiling of HFE-649 in a vertical annular minigap formed between an outer glass tube and an inner copper tube heated by a centrally located cartridge heater. Two variants of the copper heating surface were examined: a smooth surface and an enhanced surface produced by threading. The experimental measurements included fluid temperature and pressure at the inlet and outlet of the minigap, wall temperature along the test section, and the electrical parameters of the heater. The total pressure drop was analyzed using the Lockhart–Martinelli approach, with the Fanning friction factor calculated from a correlation in the literature and from an empirical relation fitted to the present dataset. Because the available pressure drop dataset is limited, the latter relation is treated here as a preliminary, geometry-specific fitting used as an auxiliary input for the present calculations rather than as a generally validated correlation. The resulting pressure drop estimates were then used to determine an effective axial velocity profile in the minigap. The thermal analysis was based on a system of steady-state energy equations for the copper tube and the fluid. The coupled inverse problems were solved using the Trefftz method, which provided two-dimensional temperature distributions in both domains and enabled the calculation of local heat transfer coefficients at the solid–fluid interface. The wall temperature results obtained using the Trefftz method were cross-checked against the Fourier-transform-based solution, with both approaches giving similar results. For the dataset considered, the empirical friction factor relation provided lower mean relative differences in pressure drop prediction than the literature relation, particularly for the smooth surface. The reconstructed wall temperature field showed good agreement with the measurements, while validation of the fluid temperature field was limited to the inlet and outlet data. For the present operating conditions, the results indicate that the threaded surface modifies local heat transfer behaviour only moderately and does not produce a pronounced overall enhancement over the entire minigap length.

Keywords: annular minigap; heat transfer; flow boiling; pressure drop; enhanced surface; Trefftz method



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1. Introduction

The continuing miniaturization of thermal systems increases the demand for compact cooling solutions capable of dissipating high heat fluxes. In this context, flow boiling in

mini- and microchannels is regarded as an efficient heat removal mechanism because it combines compact geometry with relatively high heat transfer coefficients. At the same time, the thermal benefits of such systems are accompanied by increased flow resistance, which must be considered together with heat transfer performance during design and analysis. One possible route to further intensification is the modification of the heating surface, which may alter bubble generation, liquid renewal near the wall, and the local thermal response of the system.

Boiling heat transfer in circular and rectangular minichannels has been widely studied, whereas annular minigap geometries remain less extensively documented. Previous studies on annular configurations have demonstrated the importance of gap width, orientation, heat flux, and vapour quality for the observed thermal and hydrodynamic behaviour. Experimental investigations have included subcooled flow boiling of refrigerants in narrow annuli, flow pattern studies in annular minigaps, and local heat transfer analyses in boiling flows within annular geometries [1–4]. These works confirm that annular channels constitute a distinct class of compact flow passages for which both thermal and flow resistance characteristics must be analyzed carefully.

In ref. [1], the effect of channel size on subcooled flow boiling of R-407C was investigated in horizontal annular minigaps with depths of 1 mm and 2 mm. The study emphasized that a noticeable temperature overshoot occurred at the onset of nucleate boiling (ONB), with higher heat transfer coefficients observed for smaller gaps and lower inlet subcooling. An increase in heat flux significantly enhanced heat transfer and bubble activity, whereas the effects of mass flux and saturation temperature were found to be less pronounced. Bubble visualization revealed suppression of nucleation with increasing mass flux and subcooling. The results were compared with the data for R-134a and existing correlations, and new empirical models were proposed to describe heat transfer characteristics and bubble behaviour.

Reference [2] presents an experimental investigation into the influence of gap size and inclination angle on flow pattern maps for air–water two-phase flow in an annular minigap channel. The experiments were carried out at three inclination angles (0° , 30° , and 60°) using concentric annuli with an outer diameter of 12.5 mm and inner diameters of 8, 10, and 11 mm. The results demonstrated that both the inclination angle and the gap size have a significant impact on the observed flow regimes, indicating that geometric and orientation parameters play a critical role in two-phase flow behaviour in annular configurations.

The authors of [3] describe the results of experimental studies on boiling n-hexane in annular gaps with widths of 0.5, 1.0, and 1.5 mm. Infrared thermography and Fourier-based analysis were used to determine local heat transfer coefficients. The results, plotted as functions of vapour quality up to 0.7, revealed enhanced heat transfer at higher heat fluxes and in narrower gaps. However, for the smallest gap (0.5 mm), an inverse trend was observed at high vapour qualities. Furthermore, the influence of mass flux became more significant with increasing vapour quality.

Reference [4] reports an experimental investigation of heat transfer and bubble dynamics in narrow horizontal annuli using the refrigerants R-134a, R-407C, and R-410A. A pronounced wall temperature undershoot was observed at the onset of nucleate boiling (ONB), with the magnitude decreasing in the following order: R-134a > R-407C > R-410A. Increasing the heat flux improved the boiling heat transfer coefficient for all refrigerants, with R-410A consistently achieving the highest values due to its lower surface tension. Flow visualizations showed that elevated heat flux increased buoyancy forces, resulting in larger bubble departure diameters and higher departure frequencies. Quantitative analysis of bubble dynamics based on bubble detachment diameter, bubble departure frequency,

and the active nucleation site density indicated that R-410A exhibited the smallest values among the three fluids tested.

Taken together, these studies [1–4] indicate that heat transfer in annular minigaps cannot be interpreted only as a function of heat flux. The observed behaviour is also governed by confinement, gap width, vapour quality, surface tension, reduced pressure, bubble departure dynamics, and local flow pattern development. This is why results obtained for one working fluid, orientation, or annular gap size cannot be directly transferred to another configuration. In particular, the available literature still provides limited information on how mechanically modified heating surfaces affect local HTC reconstruction in vertical annular minigaps with HFE-649.

Pressure drop constitutes an equally important aspect of minichannel performance. Earlier studies showed that under selected conditions, classical pressure drop relations may still provide reasonable predictions for smooth minichannels, whereas in two-phase flow, the predictive accuracy becomes more sensitive to geometry, diameter, surface condition, and flow regime [5–7]. For annular minigaps with boiling flow and modified wall surfaces, the available predictive tools are still limited, particularly when pressure drop estimation is further coupled with a thermal field reconstruction procedure.

In [5], experimental studies of single-phase water flow in tubular minichannels with internal diameters of 0.55 mm, 0.64 mm, and 1.10 mm showed a ($\pm 10\%$) agreement of the frictional resistance coefficient with the Hagen–Poiseuille equation for laminar flow and the Blasius equation for turbulent flow. It was found that existing formulas for smooth conventional channels can also be applied to minichannels provided that they have a smooth surface. The need to modify formulas for conventional channels was not confirmed nor was any influence of the minichannel material on flow resistance observed.

The authors of [6] present the results of an experimental investigation of two-phase pressure loss of R134a in microchannel headers using different end-cut techniques. The tests were conducted for five different refrigerant mass fluxes between $185 \text{ kg}/(\text{m}^2\text{s})$ and $785 \text{ kg}/(\text{m}^2\text{s})$, and pressure losses were recorded over the entire range of qualities, from 100% vapour to 100% liquid. A new model for estimating pressure losses for single-phase flow in the range of $\text{Re} = 800 \div 21,000$ was proposed, which showed compliance with the Churchill correlation with an error of 3.53%.

Reference [7] examined pressure drops in stainless-steel mini-pipes with internal diameters of 0.244, 0.430, and 0.792 mm, showing that single-phase flow results aligned with classical theories, with transition flow occurring at Reynolds number slightly below 2000. A new correlation was developed to estimate pressure drop in two-phase flow based on the Lockhart–Martinelli model, taking into account the effects of the tube diameter, surface tension, and Reynolds number. The new correlation predicted the experimental data with an average absolute deviation of 8.1%.

The present study addresses this issue for flow boiling of HFE-649 in a vertical annular minigap with two heating surface variants: smooth and threaded. The work combines experimental measurements with semi-analytical thermal modelling based on the Trefftz functions (T-functions), with cross-checking performed using the Fourier transform. The first objective is to assess pressure drop prediction for the present geometry and operating conditions by comparing a friction factor relation from the literature with an empirical fitting based on the present dataset. The second objective is to use the resulting pressure drop estimates to determine an effective axial velocity profile and then reconstruct the temperature fields in the copper tube and the fluid, from which local heat transfer coefficients are obtained. In this way, the paper links hydrodynamic estimation with inverse heat transfer analysis for an annular boiling configuration that has received comparatively limited attention in the literature.

The novelty of this work is not the experimental module itself, which follows the authors' earlier annular minigap studies [8], but the combined thermo-hydrodynamic reconstruction procedure applied here to HFE-649 flow boiling in a vertical annular minigap with two different copper heating surfaces. The new elements are: (i) the comparison of smooth and mechanically threaded copper heating surfaces under comparable operating conditions, (ii) the coupling of pressure drop estimation with an effective axial velocity profile used in the fluid energy equation, and (iii) the derivation and application of Trefftz functions for the fluid energy equation containing the convective term. The study therefore addresses the methodological problem of reconstructing local temperature fields and local heat transfer coefficients in an annular boiling configuration for which direct internal temperature measurements are limited.

This article is an extended and substantially revised version of the conference paper presented at the 20th Conference on Sustainable Development of Energy, Water and Environment Systems (SDEWES 2025) [9]. Compared with the conference version, the present manuscript provides an expanded methodological description; a broader discussion of model limitations, heat loss influence, and uncertainty; and a more detailed interpretation of the reconstructed temperature fields and local heat transfer coefficients.

2. Experimental Stand and Operating Conditions

The experiments were performed on a flow boiling facility designed for investigations of refrigerants in mini-scale test sections. The main loop comprised the test module, a circulation system for the working fluid, a pressure control section, a heat exchanger, flow measurement elements, pressure and temperature sensors, and a data acquisition system. Electrical power was supplied to the heater through an autotransformer, while the current and voltage drop were measured continuously during each experimental run.

In a previous study [8], the experimental setup employing a test module with a mini-scale annular gap was described in detail. The key component of the experimental stand, that is, the test module, is illustrated in Figure 1, while its cross-section is shown in Figure 2.

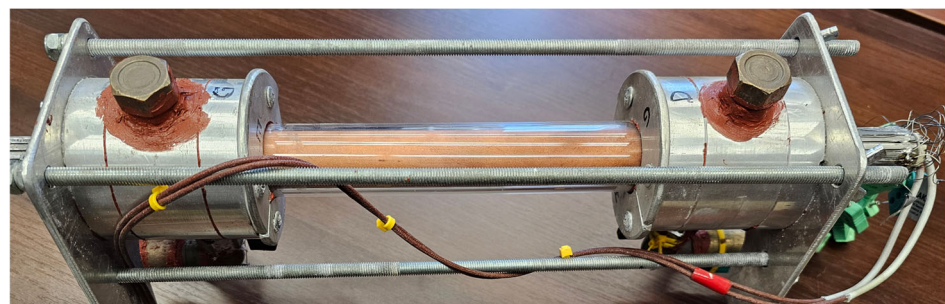


Figure 1. A view of the test module.

The experimental test module featured an annular gap with a width of 2 mm and a length of 180 mm, formed between an outer glass tube (5) and an inner copper tube (3). It was aligned coaxially with a centrally placed cartridge heater (1) of circular cross-section. HFE-649 fluid, manufactured by 3M, was used as the working medium. The test module was oriented vertically, with the fluid flowing upward. Two types of copper tube surface in contact with the working fluid were used in the experiments: one smooth and one enhanced, produced by threading.

The cartridge heater acted as the internal heat source and provided controlled heat input to the copper tube surrounding it. The heat supplied to the heating surface was determined from the measured electrical parameters of the heater. This arrangement

made it possible to analyze the thermal response of the annular minigap under gradually increasing heat flux applied under steady-state conditions.

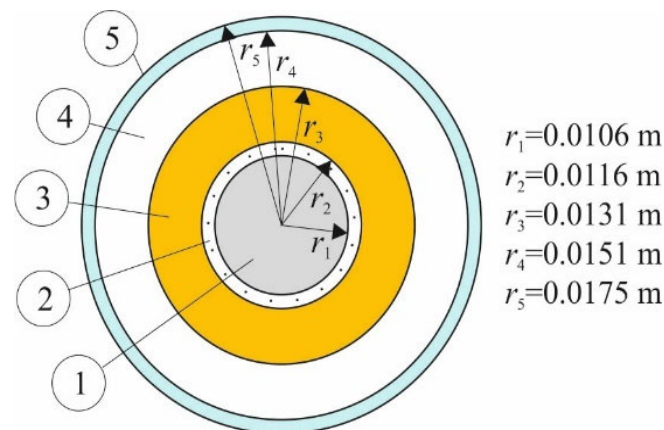


Figure 2. A cross-section of the test module: 1—cartridge heater, 2—thermally conductive filler with thermocouples, 3—copper tube, 4—minigap with HFE-649 fluid, 5—glass tube (schematic drawing, not to scale).

The heater was powered by an autotransformer that allowed for precise control of the current. The voltage drop across the heater was also recorded. On the basis of the measured electrical parameters (current and voltage drop), the heat flux delivered to the heating surface in contact with the fluid was calculated. In the present analysis, heat losses to the surroundings were neglected as a simplifying assumption because the main objective was comparative assessment of the two surface variants within the same experimental configuration. This assumption should nevertheless be regarded as a source of uncertainty in the absolute values of the calculated heat transfer coefficients.

To assess the possible influence of this assumption, an upper-bound estimate of heat losses to the surroundings was additionally performed. The loss term was approximated as the sum of natural convection and radiation losses from the external surface of the test section. The values used in this estimate were selected so as to give a conservative upper limit of the heat loss. Within the analyzed operating range, the estimated heat losses accounted for 7.36–9.26% of the electrical heat input supplied to the heater. Therefore, neglecting heat losses may shift the absolute value of the calculated heat transfer coefficient by approximately the same order of magnitude, but it does not change the comparative conclusion concerning the smooth and threaded surfaces because both surface variants were tested in the same module and under comparable thermal conditions.

At the inlet and outlet of the minigap (4), both the temperature and pressure of the HFE-649 fluid were recorded. Fluid temperature was measured using K-type thermocouples. Additional thermocouples were embedded in thermal paste between the copper tube and the heater to monitor the temperature of the heating surface. A total of 18 thermocouples (2) were distributed along the flow path at 10 mm intervals. The outer glass tube (5) enabled visual observation of the flow pattern. This arrangement provided the data required for both the pressure drop analysis and the inverse heat transfer calculations carried out in the present study.

The main measurement devices used in the experiment and their accuracies are summarized in Table 1. The accuracies were taken from the manufacturers' specifications and, where available, from additional device calibration.

During each experimental series, the electrical power supplied to the heater was gradually increased under steady-state conditions. The range of experimental parameters is listed in Table 2. For the smooth copper tube surface, the inlet overpressure ranged from

132 to 249 kPa and the outlet overpressure from 123 to 241 kPa, whereas for the enhanced surface, these ranges were 119–246 kPa and 111–238 kPa, respectively. The inlet fluid temperature was 294 K for the smooth surface and 293 K for the enhanced surface, while the outlet temperature varied from 301 to 334 K and from 301 to 330 K, respectively. The copper tube surface temperature ranged from 303 to 405 K for the smooth surface and from 303 to 425 K for the enhanced surface. The supplied current varied from 70 to 140 A for both cases, whereas the recorded voltage drop ranged from 0.81 to 1.63 V for the smooth surface and from 0.85 to 1.63 V for the enhanced surface.

Table 1. Measurement devices and measurement accuracy.

Experimental Parameter (Device, Type, Manufacturer)	Measurement Accuracy
Pressure of the fluid at the inlet to the test section, (gauge pressure meter, Cerabar S PMP71, Endress + Hauser)	$\pm 0.05\%$ of the reading, the range of $0 \div 10$ bar *
Atmospheric pressure (absolute pressure meter, A-10, Wika)	0.5% of the full scale, the range of $0 \div 2.5$ bar *
Mass flow rate (Coriolis mass flow meter, Proline Promass A100, Endress + Hauser)	$\pm 0.1\%$ of the reading, the range of $0 \div 0.125$ kg/s *
Current intensity (via ammeter)	0.1 A *
Voltage drop (via voltmeter)	$0.001\Delta U + 0.0001$ *
Temperature of the working fluid and temperature at the heater–copper tube interface (thermocouples, K-type, Czaki Thermo-Product)	± 0.9 °C [8], the range of $-40 \div 375$ °C

* According to the data provided by the manufacturers of the measuring device.

Table 2. Experimental parameters.

Measured Parameter	Range of Variability (Experiment Conducted with the Smooth Copper Tube Surface)	Range of Variability (Experiment Conducted with the Enhanced Copper Tube Surface)
Overpressure at the inlet, kPa	132–249	119–246
Overpressure at the outlet, kPa	123–241	111–238
Fluid temperature at the inlet, K	294	293
Fluid temperature at the outlet, K	301–334	301–330
Temperature of the copper tube surface, K	303–405	303–425
Current supplied to the heater, A	70–140	70–140
Voltage drop, V	0.81–1.63	0.85–1.63

The analyzed dataset therefore covers comparable operating ranges for both heating surface variants, which enables a direct comparison of the pressure drop characteristics and the reconstructed heat transfer behaviour for the smooth and threaded copper surfaces.

3. Mathematical Model

In the mathematical formulation, the vertical test module was represented as an axisymmetric system of concentric cylindrical domains. The thin layer of thermal filler and the thermocouple wires embedded in it were neglected. The model was formulated for time-independent conditions. For the present calculations, the thermophysical properties of the system components were assumed to be constant.

3.1. Pressure Drop Model and Effective Velocity Profile

Accurate determination of pressure drops in such confined geometries is essential for the optimal design and performance assessment of compact heat exchangers.

During two-phase flow in a vertical minigap, the total pressure drop $\left(\frac{\Delta p}{L}\right)_T$ was calculated by taking into account frictional, acceleration, and gravitational components [10,11], which can be written as follows:

$$\left(\frac{\Delta p}{L}\right)_T = \left(\frac{\Delta p}{L}\right)_F + \left(\frac{\Delta p}{L}\right)_A + \left(\frac{\Delta p}{L}\right)_G \quad (1)$$

The acceleration $\left(\frac{\Delta p}{L}\right)_A$ and gravitational $\left(\frac{\Delta p}{L}\right)_G$ components of the pressure loss were evaluated according to [11], whereas the frictional pressure gradient $\left(\frac{\Delta p}{L}\right)_F$ was determined using the Lockhart–Martinelli model [12] given by Equation (2):

$$\left(\frac{\Delta p}{L}\right)_F = \Phi^2 f \frac{L}{d_h} \frac{G^2(1 - X_{th})^2}{2 \rho_f} \quad (2)$$

The two-phase coefficient Φ^2 was evaluated according to [13], while the thermodynamic vapour quality X_{th} was determined from the relation adopted from [14].

In Equation (2), the Fanning friction factor was calculated using the two relations given by Equations (3) and (4) respectively

$$f Re = \frac{12}{\sqrt{\frac{r_3}{r_4}} \left(1 + \frac{r_3}{r_4}\right) \left(1 - \frac{192 \frac{r_3}{r_4} \tanh\left(\frac{\pi r_4}{2 r_3}\right)}{\pi^5}\right)} \frac{\sqrt{A}}{d_h} \quad (3)$$

$$f = 25.5 Re_f^{-0.125} \quad (4)$$

where $Re = \frac{G d_h}{\mu_f}$ and $Re_f = \frac{G d_h}{\mu_f} (1 - X_{th})$.

The correlation (3) by Lin et al. [15] was used here only as a laminar flow reference relation based on the literature because it provides a compact engineering expression for friction losses in narrow passages. Although it was originally developed for plain rectangular plate–fin cores rather than annular minigaps, it was adopted in this study as a baseline comparator, not as a geometry-exact model for the investigated test section. Equation (4) is an empirical relation fitted by the authors using the experimental data obtained for the present annular minigap. Because this fitting was developed on the basis of only 16 experimental points, it should be treated as a preliminary, dataset-specific relation intended for the present geometry and operating range. In other words, Equation (4) is used here as an auxiliary relation for the current calculations and comparison rather than as a generally validated friction factor correlation.

This distinction is important because the pressure drop part of the study serves two roles simultaneously: first, to compare the performance of a friction factor relation that is based on the literature and is dataset-specific, and second, to provide input for the thermal model. Therefore, in this paper, the pressure drop calculation should be understood primarily as a supporting step in the coupled thermo-hydrodynamic analysis. When determining the fluid velocity in the minigap, it was assumed that the fluid flow was steady state and laminar ($Re < 2100$), with only one non-zero velocity component $w(r)$ parallel to the flow direction and for $r_3 \leq r \leq r_4$, satisfying the following equation:

$$\frac{1}{r} \frac{d}{dr} \left(r \frac{dw}{dr} \right) = - \frac{1}{\mu_f} \frac{\Delta p}{L} \quad (5)$$

Solving Equation (5) yields the velocity profile given by Equation (6), where the constants A and B are determined from the homogeneous boundary conditions for $r = r_3$ and $r = r_4$ (Figure 2) expressed by Equation (7).

$$w(r) = -\frac{1}{\mu_f} \frac{\Delta P}{L} r^2 + Ar + B \ln r \quad (6)$$

$$w(r_3) = w(r_4) = 0 \quad (7)$$

For clarity, the velocity distribution obtained from Equations (5) to (7) should be interpreted as an effective axial velocity profile used in the energy equation framework adopted in this work. It is not intended to represent a complete local hydrodynamic description of the boiling two-phase structure in the annular minigap. Rather, it provides a physically consistent input required to reconstruct the temperature field and, subsequently, the local heat transfer coefficient. Such a formulation is consistent with the main aim of the present modelling approach, in which the hydrodynamic estimate of pressure drop is coupled with the inverse heat transfer analysis solved by the Trefftz method.

3.2. Governing Energy Equations and Heat Transfer Coefficient Definition

The heat transfer process in the test module was assumed to be axisymmetric and time-independent. Consequently, the temperature fields in the copper tube and in the HFE-649 fluid were described as functions of two spatial variables only: the axial coordinate z , measured along the flow direction ($0 \leq z \leq L$), and the radial coordinate r , measured across the thickness of the concentric cylindrical layers shown in Figure 2 (related to the thickness of the cartridge heater (r_1), the copper tube ($r_3 - r_1$), and the width of the minigap ($r_4 - r_3$)). In the mathematical model, it was assumed that the temperatures of the copper tube (T_C) and the HFE-649 fluid (T_f) satisfy the energy equation given by Equation (8):

$$\lambda_i \mathcal{L} T_i = D c_{p,i} \rho_i w(r) \frac{\partial T_i}{\partial z} \quad (8)$$

In this equation, the differential operator $\mathcal{L} = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial}{\partial r} \right) + \frac{\partial^2}{\partial z^2}$, and the values of index i and constant D corresponding to the respective temperature fields are listed in Table 3. For the fluid domain, the convective term includes the axial velocity $w(r)$ determined previously from Equation (6).

Table 3. Values of index i and constant D .

Refers to the Temperature	Index i	Constant D
Copper tube	C	0
Fluid HFE-649	f	1

It should be emphasized that the adopted velocity profile and the fluid energy equation are not intended to represent a complete mechanistic model of the local two-phase boiling structure. The formulation does not resolve vapour–liquid interfacial dynamics, eddy viscosity, turbulent Prandtl number effects or local flow pattern transitions. Instead, it is used as an effective, reconstruction-oriented description that enables the temperature field in the fluid domain to be coupled with the measured wall temperature data and with the pressure drop estimate. Consequently, the obtained heat transfer coefficient should be interpreted as a local coefficient reconstructed within the adopted inverse framework, not as a universal predictive boiling correlation.

The adopted formulation reflects the coupled character of the problem: Heat is conducted through the copper tube and simultaneously transported and redistributed in

the fluid flowing through the annular minigap. For the copper tube, it was assumed, similarly to [8], that the temperature and heat flux at the contact surface with the heater are known from the experiment, while the remaining outer boundaries of the tube are thermally insulated.

For the HFE-649 fluid, the following boundary conditions were assumed: The fluid temperature at the inlet and outlet of the minigap is known, and the fluid remains in perfect thermal contact with the copper tube, which means continuity of both temperature and heat flux at the solid–fluid interface. Under these assumptions, the system of differential Equation (8), supplemented by the adopted boundary conditions, leads to two coupled inverse Cauchy-type heat transfer problems [16]: one formulated in the copper tube and the other in the fluid domain.

This formulation is consistent with the measurement layout used in the experiment. The available temperature data are imposed at selected boundaries, whereas the internal temperature fields must be reconstructed from the governing equations and the interfacial coupling conditions.

Knowledge of the functions T_C and T_f allows the determination of the local heat transfer coefficients at the interface between the copper tube and the fluid based on the Robin boundary condition.

$$\alpha(z) = \frac{-\lambda_c \frac{\partial T_c}{\partial r}}{T_c(r_3, z) - T_{f,ave}(z)} \quad (9)$$

where the reference temperature $T_{f,ave}$ was calculated as in [14].

In this way, the local heat transfer coefficient is not prescribed a priori but obtained as a result of the coupled temperature field reconstruction.

It should be emphasized that within the present framework, the calculated heat transfer coefficient is an outcome of the semi-analytical inverse procedure based on the measured boundary data and the adopted model assumptions. Therefore, its accuracy depends not only on the mathematical solution method but also on the quality of the experimental input data and on the validity of the simplifying assumptions used in the formulation of the problem.

4. Numerical Method

The coupled inverse problems formulated for the copper tube and the fluid domain were solved using the Trefftz method [8,17]. In ref. [8], the T-functions for the Laplace equation in polar coordinates were introduced. This paper develops this concept by deriving new T-functions for the energy equation containing the convective term associated with the axial velocity profile $w(r)$. These functions were derived using the inverse operator method, which enables the recursive determination of their analytical form for a prescribed differential operator. The proposed approach allows the direct construction of functions that satisfy the considered differential equations in a strict sense, without introducing additional approximations. For the resulting functions and their derivatives, the fundamental analytical properties were determined and formally established, including the corresponding recurrence relations.

To verify the wall temperature calculations obtained using the T-function approach, the Fourier transform was applied to determine the temperature of the copper tube. The method used to determine the fluid temperature depended on the prevailing heat transfer mechanism.

4.1. Trefftz Functions for the Copper Tube Problem

In ref. [8], a methodology was presented to determine the T-function for the Laplace equation in the form (8) when $D = 0$:

$$\mathcal{L}T = 0 \quad (10)$$

The T-functions for Equation (10) can be written in the following form:

$$g_n(r, z) = \sum_{k=0}^{\lfloor \frac{n}{2} \rfloor} \frac{(-1)^k}{(k!)^2 (n-2k)!} \left(\frac{r}{2}\right)^{2k} z^{n-2k} \quad (11)$$

$$h_n(r, z) = g_n(r, z) \cdot \ln r - \sum_{k=0}^{\lfloor \frac{n}{2} \rfloor} \frac{(-1)^n H_k}{(k!)^2 (n-2k)!} \left(\frac{r}{2}\right)^{2k} z^{n-2k} \quad (12)$$

where H_k denotes harmonic numbers, and the $\lfloor n/2 \rfloor$ denotes the floor function.

These functions were used to approximate the temperature field in the copper tube, where heat transfer is governed by conduction only. Their selection is consistent with the mathematical structure of the copper tube problem and with the axisymmetric geometry of the analyzed test module.

4.2. Trefftz Functions for the Fluid Energy Equation

For the fluid domain, the governing equation differs from the Laplace equation because it includes the convective contribution associated with the fluid motion. Therefore, a separate set of T-functions was introduced using the operator method for the energy equation given in Equation (13), in which the velocity field is the same as that defined earlier in Equation (6):

$$\mathcal{L}T = w(r) \frac{\partial T}{\partial z} \quad (13)$$

First, the inverse operator $\mathcal{L}^{-1}$ was defined for algebraic expressions as shown in Equations (14)–(17). By combining these relations with the T-functions corresponding to the Laplace equation, a new class of functions satisfying Equation (13) was derived, in the forms presented in Equation (19). Their key properties are summarized in Table 4.

Table 4. Properties of functions $g_n(r, z)$, $g_{n,i}(r, z)$, and $u_n(r, z)$ and their partial derivatives.

Function $f_n(r, z)$	Function $u_n(r, z)$	Function $f_{n,i}(r, z)$
Satisfy the Laplace equation $\mathcal{L}f_n = 0$	Satisfy the energy equation $\mathcal{L}u_n = w(r) \frac{\partial u_n}{\partial z}$	
$\frac{\partial f_0}{\partial z} = 0$ $\frac{\partial f_n}{\partial z} = f_{n-1}$	$\frac{\partial u_0}{\partial z} = 0$ $\frac{\partial u_n}{\partial z} = u_{n-1}$	$\frac{\partial f_{0,0}}{\partial z} = 0$ $\frac{\partial f_{n,i}}{\partial z} = f_{n-1,i}$ for $i = 1, 2, \dots, n$
For a fixed point (r, z) at $n \rightarrow \infty$, we have		
$\lim f_n = 0$	$\lim u_n = 0$	$\lim f_{n,i} = 0$

For $k = 0$ and $k = 1$, the following relation is obtained:

$$\mathcal{L}^{-1}(z^k r^n \ln^m r) = z^k \int \frac{1}{r} \left(\int r^{n+1} \ln^m r \, dr \right) dr \quad (14)$$

where k , n , and m are natural numbers.

Since $\mathcal{L}$ is a linear differential operator, by calculating $\mathcal{L}$ from the expression $z^k \mathcal{L}^{-1}(r^n \ln^m r)$, for $k = 2, 3, \dots$, it is obtained that:

$$\begin{aligned} \mathcal{L} \left[z^k \mathcal{L}^{-1}(r^n \ln^m r) \right] &= z^k \mathcal{L} \left[\mathcal{L}^{-1}(r^n \ln^m r) \right] + k(k-1) z^{k-2} \mathcal{L}^{-1}(r^n \ln^m r) \\ &= z^k r^n \ln^m r + k(k-1) z^{k-2} \mathcal{L}^{-1}(r^n \ln^m r) \end{aligned} \quad (15)$$

and

$$z^k r^n \ln^m r = \mathcal{L} \left[z^k \mathcal{L}^{-1}(r^n \ln^m r) \right] - k(k-1) z^{k-2} \mathcal{L}^{-1}(r^n \ln^m r) \quad (16)$$

The final formula was obtained by applying the operator $\mathcal{L}^{-1}$ to both sides of Equation (16).

$$\mathcal{L}^{-1}(z^k r^n \ln^m r) = \begin{cases} z^k \int \frac{1}{r} \left(\int r^{n+1} \ln^m r \, dr \right) dr & \text{for } k = 0, 1 \\ z^k \mathcal{L}^{-1}(r^n \ln^m r) - k(k-1) \mathcal{L}^{-1} \left[z^{k-2} \mathcal{L}^{-1}(r^n \ln^m r) \right] & \text{for } k = 2, 3, \dots \end{cases} \quad (17)$$

Using Equation (17) and the T-functions given by Equation (11), new T-functions were defined to satisfy Equation (13) in the following form:

$$u_n(r, z) = \sum_{i=0}^n g_{n,i}(r, z) \quad (18)$$

and

$$g_{n,i}(r, z) = \begin{cases} g_n(r, z) & \text{for } i = 0 \\ \mathcal{L}^{-1} \left(w(r) \frac{\partial g_{n,i-1}}{\partial z} \right) & \text{for } i = 1, 2, \dots, n \end{cases} \quad (19)$$

The analytical construction of these functions is an important element of the proposed approach, as it provides basis functions that satisfy the fluid energy equation exactly throughout the domain rather than approximately at discrete points. In this sense, the method also preserves the main advantage of Trefftz formulations for the fluid problem involving convection.

As noted in the derivation, the function defined by Equation (18) is a polynomial of degree n with respect to the variable z . For the following function, the highest power of z is $n - i$:

$$g_{n,i}(r, z) = \mathcal{L}^{-1} \left(w(r) \frac{\partial g_{n,i-1}}{\partial z} \right) \quad (20)$$

On the other hand, the following function depends solely on the radial coordinate r :

$$g_{n,n}(r, z) = \mathcal{L}^{-1} \left(w(r) \frac{\partial g_{n,n-1}}{\partial z} \right) \quad (21)$$

From the above considerations and the properties of the function $g_n(r, z)$, as given in [8], the formulas presented in Table 4 follow.

The proof that the functions defined by Equation (18) satisfy Equation (13) follows from the transformations shown in Equations (22)–(24). Using the properties of the functions $g_n(r, z)$ and $g_{n,i}(r, z)$, it can be shown that the functions $u_n(r, z)$, defined by formula (18), also satisfy Equation (13). Substituting expression (18) into Equation (13), we get:

$$\mathcal{L}(u_n(r, z)) = \mathcal{L} \left(\sum_{i=0}^n g_{n,i}(r, z) \right) = w(r) \sum_{i=0}^n \frac{\partial g_{n,i}(r, z)}{\partial z} \quad (22)$$

Since the operator $\mathcal{L}$ is linear, the left-hand side of Equation (22) can be written in the following form (for clarity, the arguments r and z are omitted in the transformations below):

$$\mathcal{L}(\sum_{i=0}^n g_{n,i}) = \sum_{i=0}^n \mathcal{L}g_{n,i} = \mathcal{L}\left(g_n + \sum_{i=1}^n \mathcal{L}^{-1}\left(w \frac{\partial g_{n,i-1}}{\partial z}\right)\right) = w \sum_{i=1}^n \frac{\partial g_{n,i-1}}{\partial z} \quad (23)$$

Transforming the expression on the right-hand side of Equation (22), we obtain:

$$w \sum_{i=0}^n \frac{\partial g_{n,i}}{\partial z} = w \sum_{i=0}^{n-1} \frac{\partial g_{n,i}}{\partial z} = w \sum_{i=1}^n \frac{\partial g_{n,i-1}}{\partial z} \quad (24)$$

which means that the functions $u_n(r, z)$ satisfy Equation (13). Formulas (18) and (19) are valid for any velocity profile $w(r)$ provided that the inverse operator $\mathcal{L}^{-1}$ is specified for all components of the expression $z^k r^n (w(r))^m$.

In practical terms, this means that the newly defined T-functions can be used as admissible basis functions for reconstructing the fluid temperature field in the annular minigap under the steady-state conditions.

4.3. The Trefftz Method

The system of energy equations, together with the adopted boundary conditions, defines two Cauchy-type inverse problems for the copper tube and the fluid, respectively. The T-functions defined by Formulas (11), (12) and (18) are used to determine the two-dimensional temperature distribution of the heating surface and the fluid flowing through the minigap, respectively. The unknown functions T_C and T_f are expressed as linear combinations of the appropriate functions; i.e.,

$$T_C(r, z) = \sum_{n=0}^{N_1} a_n g_n(r, z) + \sum_{n=0}^{N_2} b_n h_n(r, z) \quad \text{for } r_2 \leq r \leq r_3 \text{ and } 0 \leq z \leq L \quad (25)$$

$$T_f(r, z) = \sum_{n=0}^{N_3} c_n u_n(r, z) \quad \text{for } r_3 \leq r \leq r_4 \text{ and } 0 \leq z \leq L \quad (26)$$

The unknown coefficients a_n , b_n , and c_n are determined based on the assumed boundary conditions analogously to [8].

The resulting temperature fields are two-dimensional, continuous, and differentiable, and they satisfy the governing differential Equation (8) exactly within the corresponding domains. The boundary conditions are satisfied approximately. Once the temperature fields of the copper tube and the fluid are obtained, the local heat transfer coefficient can be calculated from the Robin condition introduced in Equation (9).

This numerical procedure links the experimentally prescribed boundary data, the pressure-drop-based effective velocity profile, and the analytical properties of the Trefftz basis functions into one coupled reconstruction framework.

4.4. The Fourier Transform

The two-dimensional temperature field of the copper tube was determined using the Fourier transform, with the integral transform and its inversion for the function $T_C(r, z)$ in the finite interval $0 \leq z \leq L$ given by the following formulas:

$$\mathcal{F}\{T_C(r, z)\} = \frac{2}{L} \int_0^L T_C(r, z) \cos\left(\frac{n\pi z}{L}\right) dz \quad (27)$$

$$T_C(r, z) = \frac{2}{L} \sum_{n=0}^{\infty} \mathcal{F}\{T_C(r, z)\} \cos\left(\frac{n\pi z}{L}\right) \quad (28)$$

Finally, in a finite ring, the solution is expressed as:

$$T_C(r, z) = \sum_{n=0}^{\infty} \left(A_n I_0\left(\frac{n\pi r}{L}\right) + B_n K_0\left(\frac{n\pi r}{L}\right) \right) \cos\left(\frac{n\pi z}{L}\right) \quad (29)$$

The unknown coefficients A_n and B_n were determined following the procedure described in [17], assuming that the temperature and heat flux are specified at the interface with the cartridge heater. In this approach, the reference temperature of the fluid used to determine the heat transfer coefficient is defined according to the heat transfer mechanism:

- For single-phase flow and in the subcooled boiling region, the fluid temperature varies linearly along the minichannel from the inlet to the outlet;
- For convective boiling in the saturated boiling region, the fluid temperature is taken as the local saturation temperature.

5. Results and Discussion

5.1. Input Data Preparation and Wall Temperature Approximation

The calculations were performed for both heating surface variants, i.e., the smooth and enhanced copper tube, using eight analyzed heat flux values ranging from 4.55 to 18.33 kW/m² at a mass flux of 18.82 kg/(m²s). The corresponding ranges of the measured operating parameters are summarized in Table 2.

Before solving the inverse heat transfer problems, the temperature data measured at the heater–copper tube interface were approximated by a polynomial in order to reduce the influence of local measurement scatter on the subsequent calculations.

Better approximations were obtained for the smooth copper surface, for which the average coefficient of determination was 0.87. For the enhanced surface, the corresponding value was 0.79. Figure 3 presents the results of the approximation procedure for both analyzed heating surfaces, where the measured values are indicated by points and their approximation by a dashed line. The lower quality of approximation obtained for the threaded surface is consistent with the more complex local thermal response expected for a geometrically modified wall.

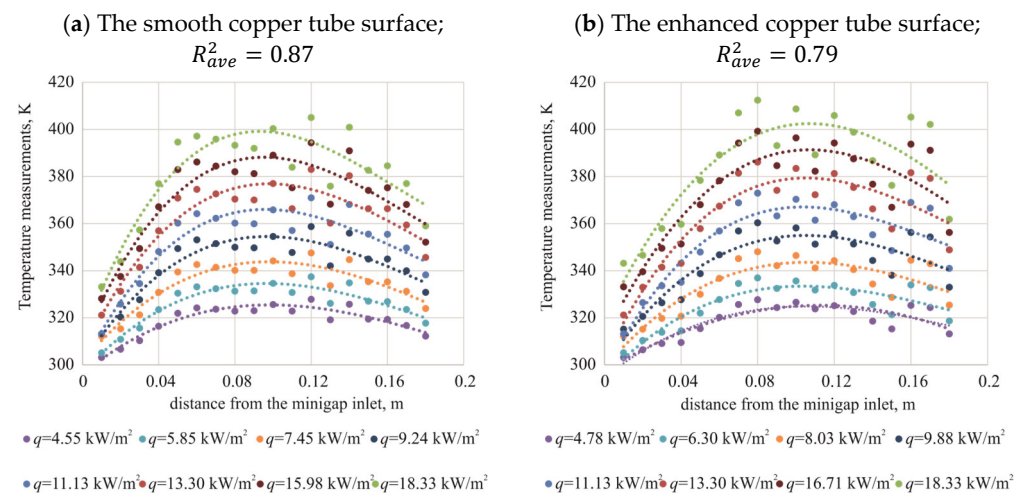


Figure 3. Temperature measurements at the heater–copper tube interface for (a) the smooth copper tube surface and (b) the enhanced copper tube surface.

The strong axial variation in the temperature measured at the heater–copper tube interface should not be interpreted as a measurement artefact alone. It results from the combined effect of finite heater length, axial conduction in the copper tube, local thermal contact between the cartridge heater, thermal filler and copper tube, and heat removal by the fluid flowing upward through the annular minigap. The lower temperatures observed near the inlet and outlet are mainly associated with end effects. In these regions, the thermal field is affected by the proximity of the inlet/outlet sections and by the axial redistribution

of heat in the solid wall. Therefore, the first and last measurement locations are more sensitive to boundary conditions than the central part of the minigap.

The approximation step is therefore treated here as a part of input data preparation for the inverse analysis rather than as a result in itself. Its purpose is to provide smooth boundary data for the copper tube problem while preserving the experimentally observed axial temperature trend.

5.2. Assessment of Pressure Drop Predictions

The total pressure drop was estimated using the two Fanning friction factor relations listed in Table 3. Tables 5 and 6 compare the experimentally measured pressure drops with the values calculated using Equation (3) and Equation (4), respectively, together with the corresponding relative differences (RDs). Table 5 refers to the smooth copper tube surface, whereas Table 6 presents the results for the enhanced (threaded) surface.

Table 5. Experimental and calculated total pressure drops using Equations (3) and (4), together with the corresponding RD, for various heat flux values. The results correspond to the experiment carried out with the smooth copper tube surface.

1	2	3	4	5	6
Heat Flux q	Experimental Pressure Drop	Calculated Pressure Drop, with the Use of Formula (3)	RD for Data from Columns 2 and 3	Calculated Pressure Drop, with the Use of Formula (4)	RD for Data from Columns 2 and 5
kW/m ²	Pa	Pa	%	Pa	%
4.55	9000	5916.76	34.25%	5869.45	34.78%
5.85	9000	7291.58	18.98%	7302.8	18.86%
7.45	9000	8829.22	1.89%	8959.03	0.46%
9.24	10,000	9411.67	5.88%	9712.93	2.87%
11.13	9000	9046.69	0.51%	9481.08	5.35%
13.30	9000	8133.21	9.63%	8699.77	3.34%
15.98	8000	6412.39	19.84%	7152.64	10.59%
18.33	8000	6935.11	13.31%	7912.81	1.09%

For the smooth surface, Equation (4) reduced the relative difference in six out of eight analyzed cases. For the enhanced surface, a reduction was obtained in four out of eight cases. The mean RD decreased from 13.04% to 9.67% for the smooth surface and from 13.74% to 12.89% for the enhanced surface when Equation (4) was used instead of Equation (3). This means that the improvement introduced by Equation (4) is clearly more pronounced for the smooth surface than for the threaded one.

At the same time, the results show that the predictive advantage of Equation (4) remains moderate for the enhanced surface and is not uniform over the whole range of heat fluxes. The largest discrepancies between the experimental and calculated pressure drops still occur at the lowest and highest heat fluxes. For this reason, Equation (4) should not be presented as a generally validated friction factor correlation but, rather, as a preliminary dataset-specific fitting that improves agreement for the present annular minigap dataset and is sufficient for the auxiliary calculations performed in this study.

The pressure drops listed in column 5 of Tables 5 and 6, obtained using Equation (4), were subsequently used to determine the fluid velocity profile in the minigap according to Equation (6). For both surface types, the refrigerant velocity ranged from 0.009 m/s to 0.014 m/s, with an average value of approximately 0.012 m/s. The maximum value of 0.014 m/s was obtained for heat fluxes of 9.24 kW/m² for the smooth surface and

9.88 kW/m² for the enhanced surface. These values were then used to define the effective axial velocity profile required in the fluid energy equation.

Table 6. Experimental and calculated total pressure drops using Equations (3) and (4), together with the corresponding RD, for various heat flux values. The results correspond to the experiment carried out with the enhanced copper tube surface.

1	2	3	4	5	6
Heat Flux q	Experimental Pressure Drop	Calculated Pressure Drop, with the Use of Formula (3)	RD for Data from Columns 2 and 3	Calculated Pressure Drop, with the Use of Formula (4)	RD for Data from Columns 2 and 5
kW/m ²	Pa	Pa	%	Pa	%
4.78	8000	6015.19	24.81%	5985.36	25.18%
6.30	9000	7984.69	11.28%	8033.61	10.74%
8.03	8000	8653.24	8.17%	8799.65	10.00%
9.88	9000	9128.59	1.43%	9423.28	4.70%
11.13	9000	8770.51	2.55%	9150.66	1.67%
13.30	8000	8228.39	2.85%	8777.43	9.72%
16.71	8000	6184.34	22.70%	6892.95	13.84%
18.33	8000	5112.39	36.10%	5816.95	27.29%

The above results should be interpreted with caution. Equation (4) was fitted to a limited dataset obtained for the present annular minigap geometry and a narrow operating range; therefore, it should be treated as a preliminary dataset-specific relation rather than a general friction factor correlation. Moreover, the resulting velocity profile is used here as an effective input to the thermal model, not as a complete local hydrodynamic description of the boiling two-phase flow.

5.3. Reconstructed Temperature Fields

In the next step, the temperatures of the copper tube and the fluid were determined using the Trefftz method, assuming the same number of T-functions in the approximations given by Equations (18) and (19). Figure 4 presents the RD between the calculated and measured wall temperature for the enhanced surface as a function of the distance from the minigap inlet, as well as the mean relative differences (MRD) as a function of heat flux for both analyzed heating surfaces.

The maximum MRD was 0.78% for the smooth heating surface and 0.93% for the enhanced surface. Higher local RD values were observed for the enhanced surface, with the largest discrepancies appearing at the initial measurement locations. This indicates that the threaded surface produces a somewhat more demanding reconstruction problem, but the overall agreement between calculation and measurement remains very good for both surface variants.

Considering the refrigerant temperature, MRD could be determined only for the inlet and outlet temperatures of the minigap because these were the only fluid temperature values directly available from the experiment. For both heating surfaces, the MRD did not exceed 2% for the inlet temperature and 1% for the outlet temperature. The largest differences were observed at the highest heat fluxes. Accordingly, the reconstructed fluid temperature field should be regarded as consistent with the available boundary measurements, although its internal validation remains limited by the measurement layout.

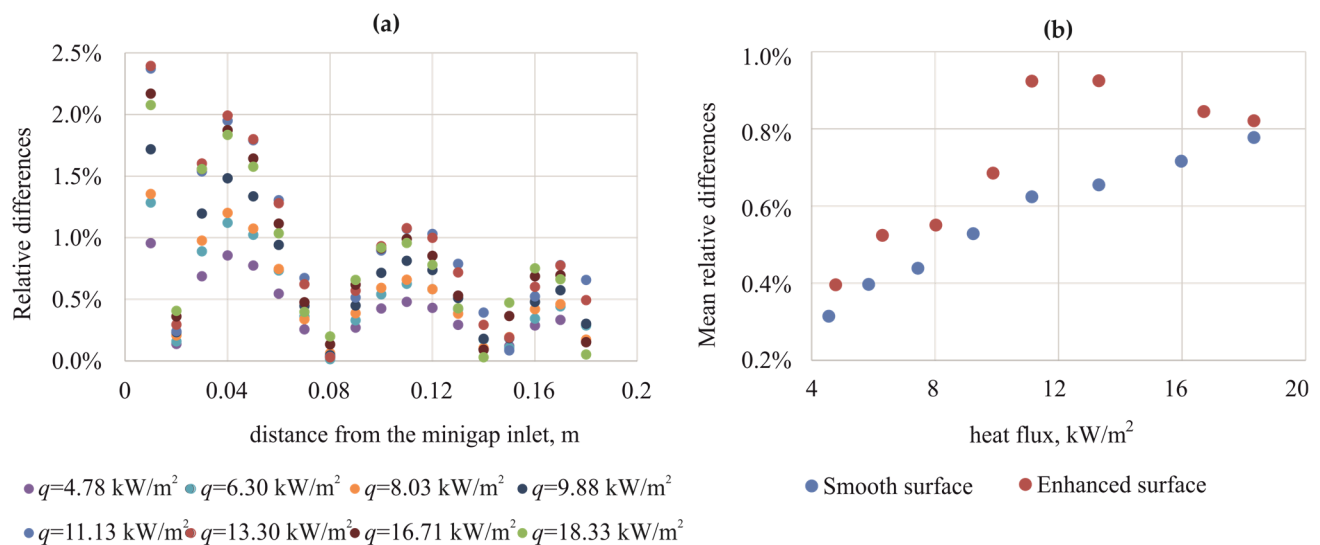


Figure 4. (a) Relative differences for the enhanced-surface temperature as a function of distance from the minigap inlet; (b) mean relative differences for both analyzed surfaces as a function of heat flux.

Figure 5 shows representative two-dimensional temperature distributions in the solid and fluid domains for the enhanced-surface case.

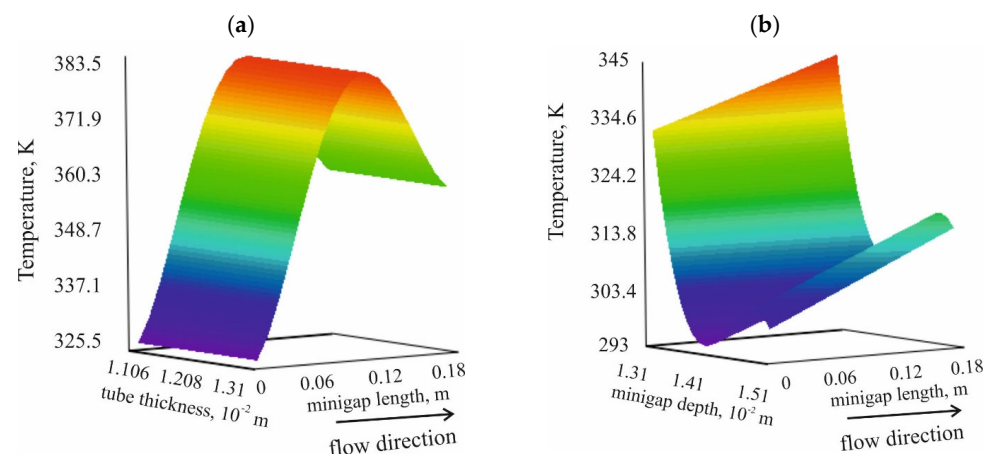


Figure 5. Two-dimensional temperature distributions in the solid and fluid domains for the enhanced-surface case: (a) copper tube; (b) HFE-649 fluid, for $q = 13.30 \text{ kW/m}^2$. The remaining operating conditions are listed in Table 2.

For the representative case shown in Figure 5, it was observed that the wall temperature initially increases along the minigap length, reaches a maximum, and then gradually decreases in the downstream part of the channel. This behaviour reflects the changing intensity of heat removal from the heated wall as the fluid is progressively heated and boiling becomes more pronounced. The fluid temperature is highest in the near-wall region, i.e., directly at the contact with the heated surface, and decreases toward the channel core because of the transverse temperature gradient. At the same time, the fluid temperature increases in the axial direction as the refrigerant absorbs heat while flowing through the minigap.

5.4. Local Heat Transfer Coefficient Distributions

Figures 6 and 7 present the values of local heat transfer coefficients as a function of the distance from the inlet to the minigap. The Fanning friction factor used in these calculations was obtained from Equation (4).

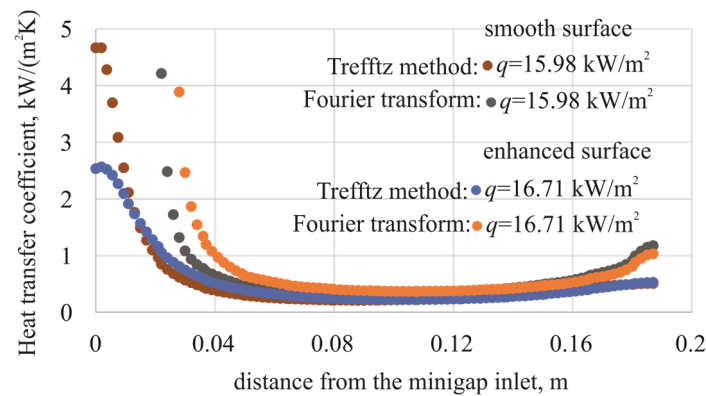


Figure 6. Comparison of the local heat transfer coefficient for smooth and enhanced copper tube surfaces during convective boiling, obtained with the T-function and the Fourier transform. The corresponding operating ranges are summarized in Table 2.

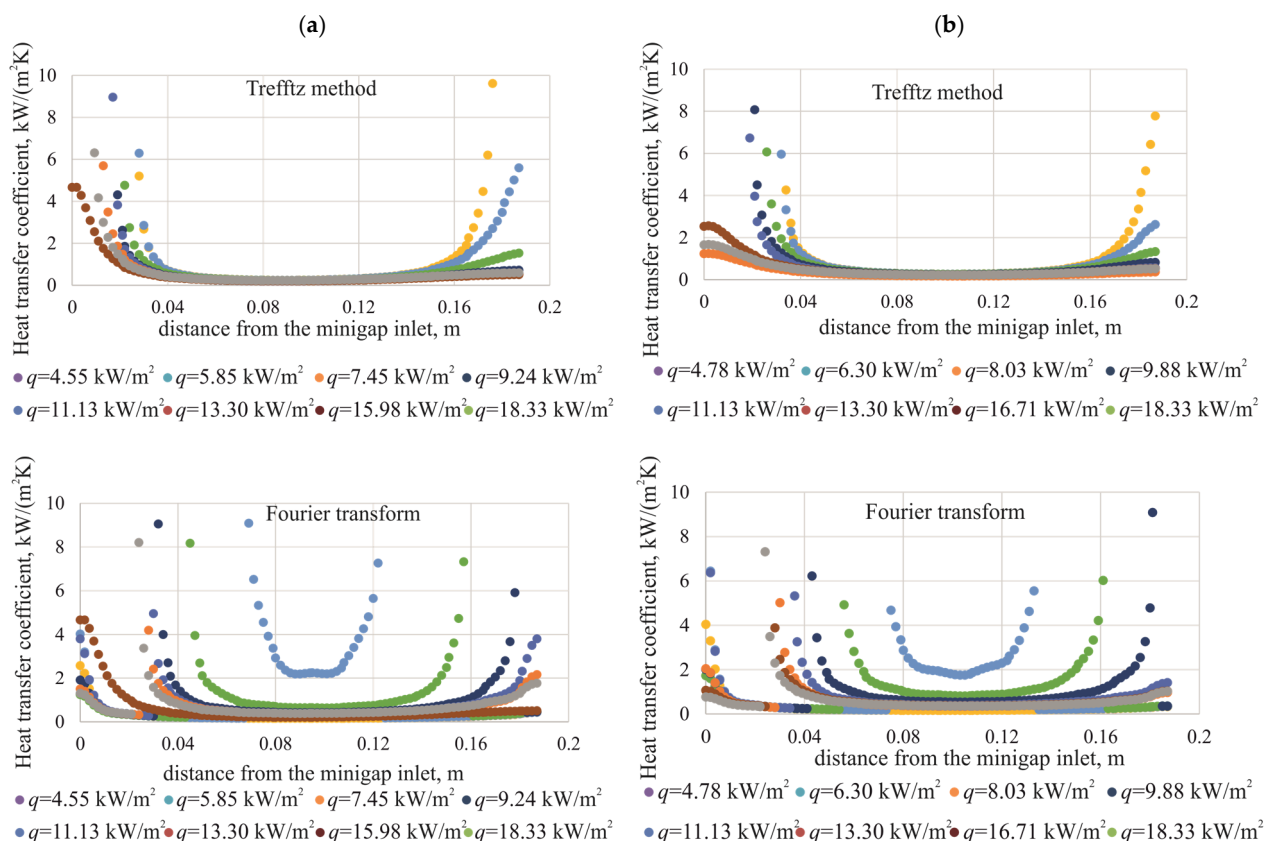


Figure 7. Local heat transfer coefficient versus distance from the minigap inlet for (a) the smooth heating surface and (b) the enhanced heating surface. The corresponding operating ranges are summarized in Table 2.

For convective boiling, Figure 6 presents representative local heat transfer coefficient distributions for the smooth ($q = 15.98 \text{ kW/m}^2$) and enhanced ($q = 16.71 \text{ kW/m}^2$) copper tube surfaces obtained using both calculation methods, i.e., the T-function method and the Fourier transform. Both methods yielded similar results, particularly in the middle section of the channel. Similar trends were observed for the remaining analyzed heat flux values shown in Figure 7, except for the case of $q = 5.85 \text{ kW/m}^2$.

Figure 7 presents the local heat transfer coefficient as a function of the distance from the annular gap inlet, calculated for both types of heating surfaces using the T-function and the Fourier transform.

When analyzing the results shown in Figure 7, it was observed that regardless of the calculation method, in the inlet region extending approximately to the midpoint of the minigap, the wall temperature increases whereas the local heat transfer coefficient decreases along the flow direction. This behaviour is consistent with the development of the thermal field before boiling becomes more intense in the downstream region.

As boiling intensifies, the observed reduction in wall temperature is consistent with increasing nucleation activity and enhanced heat removal from the wall, which is accompanied by an increase in the heat transfer coefficient. In the downstream section of the minigap, a further decrease in the heating surface temperature is observed, while the heat transfer coefficient continues to increase. A similar overall trend was obtained for both the smooth and enhanced heating surfaces.

The comparison of the two surfaces shows that the general axial distributions of the heat transfer coefficient are similar over most of the heating length. The effect of surface enhancement is therefore moderate under the present operating conditions. Slightly higher values for the enhanced surface can be observed in selected regions, especially near the minigap entrance for lower heat fluxes and, in some cases, in the outlet region. However, the threaded surface does not produce a pronounced overall increase in heat transfer coefficient over the entire test section.

For this reason, the present results support the conclusion that the threaded surface modifies local heat transfer behaviour only to a limited extent rather than fundamentally changing the thermal response of the annular minigap.

The moderate influence of threading can be explained by the balance between several competing effects. The threaded surface increases the geometrical complexity of the wall and may locally promote vapour trapping or earlier activation of nucleation sites. However, under the present low-mass-flux upward-flow conditions, the overall heat transfer is strongly governed by the development of the boiling flow, liquid replenishment near the wall, and vapour removal along the annular gap. In such a confined geometry, a mechanically threaded surface does not automatically lead to a proportional increase in the active nucleation site density or to a stable improvement in fluid–heated wall contact. Moreover, local vapour accumulation in the grooves may partially reduce the benefit of the increased surface area. As a result, the threaded surface modifies the local HTC distribution, but the net enhancement averaged over the whole minigap remains limited.

For the investigated operating conditions, the recommended design-oriented interpretation is based on averaged HTC values evaluated over the central part of the minigap (0.02–0.16 m), excluding the inlet and outlet regions.

The largest local values of the heat transfer coefficient occur near the inlet and outlet regions. These values should not be treated as fully developed, design-representative HTC values. They are local reconstruction results obtained in zones where the inverse problem is most sensitive to boundary conditions, axial conduction, and the adopted reference fluid temperature. In particular, when the temperature difference used in the Robin condition becomes small, even moderate uncertainty in wall or reference fluid temperature may lead to a large local change in the calculated heat transfer coefficient. Additionally, the identification of the temperature fields in both the copper tube and the fluid required the solution of two consecutive inverse heat transfer problems. This sequential reconstruction procedure further increased the sensitivity of the obtained results to the imposed boundary data, particularly in the case of the Fourier transform approach, which was reflected most clearly in the HTC distributions near the minigap outlet.

For engineering design purposes, the most reliable values are therefore not the extreme local peaks but spatially averaged HTC values obtained over the central part of the minigap, where both the Trefftz-based and Fourier-transform-based solutions show the best mutual

agreement. The end regions should be used mainly for qualitative interpretation of boiling incipience and boundary effects, not for selecting a representative heat transfer coefficient for compact heat exchanger design.

5.5. Uncertainty Analysis

The relative errors of the heat transfer coefficient were calculated using the standard law of uncertainty propagation from the following formula:

$$\Delta\alpha = \frac{1}{\alpha(z)} \left[\left(\frac{\partial\alpha}{\partial\lambda_C} \Delta\lambda_C \right)^2 + \left(\frac{\partial\alpha}{\partial\frac{\partial T_c}{\partial r}} \Delta\frac{\partial T_c}{\partial r} \right)^2 + \left(\frac{\partial\alpha}{\partial T_c} \Delta T_c \right)^2 + \left(\frac{\partial\alpha}{\partial T_{f,ave}} \Delta T_{f,ave} \right)^2 \right]^{0.5} \quad (30)$$

where:

$\Delta\lambda_C = 0.1 \text{ W/(mK)}$ —the uncertainty of the thermal conductivity of the copper tube;

$\Delta\frac{\partial T_c}{\partial r} = \left| \frac{\partial^2 T_c}{\partial r \partial z} \Delta z \right|$ —the uncertainty in the copper tube temperature gradient;

$\Delta T_c = \left[(\Delta T_{meas})^2 + \left[\frac{\partial T_c}{\partial z} \Delta z \right]^2 \right]^{0.5}$ —the uncertainty in the copper tube temperature;

$\Delta T_{meas} = 0.9 \text{ K}$ —the uncertainty in the heated tube temperature (see Table 1) obtained with additionally calibrated K-type thermocouples [8];

$\Delta T_{f,ave} = \left[(\Delta T_f)^2 + \left[\frac{\partial T_{f,ave}}{\partial z} \Delta z \right]^2 \right]^{0.5}$ —the uncertainty in the fluid temperature calculated using the Trefftz method;

$\Delta T_{f,ave} = \left[(\Delta T_f)^2 + [\Delta T_{f,ave}]^2 \right]^{0.5}$ —the uncertainty in the fluid temperature calculated using the Fourier transform;

$\Delta T_f = 0.9 \text{ K}$ —the uncertainty in fluid temperature measurements obtained with additionally calibrated K-type thermocouples [8];

$\Delta z = 10^{-4} \text{ m}$ —the uncertainty in the temperature measurement location.

The relative errors of the HTC were determined for individual measurement points, and subsequently, the mean relative error (MRE) was calculated based on these data. The analysis of the MRE of the HTC demonstrated a significant influence of both the applied calculation method and the heater surface geometry on the accuracy of the obtained results. For the enhanced surface, the Fourier transform method provided average relative HTC errors ranging from 6.87% to 20.48%, whereas the Trefftz method for the same surface resulted in errors ranging from 3.73% up to 65.54%. In the case of the smooth surface, the Trefftz method yielded more stable results, with MRE varying between 4.38% and 31.69%. The analysis also revealed distinct differences between subcooled boiling and saturated nucleate boiling. For saturated nucleate boiling, the lowest relative HTC errors were obtained, ranging from 3.09% to 8.68% for the Trefftz method and from 6.74% to 29.41% for the Fourier transform method, indicating a more stable heat transfer process. In contrast, during subcooled boiling, significantly larger data scatter and maximum pointwise relative errors reaching up to 140% were observed. This phenomenon mainly occurred near the channel inlet, where flow and heat transfer conditions were the least stabilized and temperature gradients were the highest. Moreover, it was observed that the MRE generally decreased with increasing heat flux, indicating improved stability of the boiling process at higher thermal loads.

6. Conclusions

This study investigated flow resistance and heat transfer during flow boiling of HFE-649 in a vertical annular minigap with a smooth and an enhanced (threaded) copper heating surface. The analysis, combining experimental measurements with a semi-analytical

reconstruction procedure based on the Trefftz method, was compared with results obtained using the Fourier transform.

The main conclusions of the present study can be summarized as follows:

- The empirical Fanning friction factor relation given by Equation (4) provided lower mean relative differences in pressure drop prediction than the literature relation of Equation (3), particularly for the smooth surface. The mean RD decreased from 13.04% to 9.67% for the smooth surface and from 13.74% to 12.89% for the enhanced surface. However, because Equation (4) was fitted using only 16 experimental points, it should be treated as a preliminary dataset-specific relation rather than as a generally validated friction factor correlation.
- The Trefftz-based approach enabled the solution of two coupled inverse Cauchy-type heat transfer problems formulated for the copper tube and the fluid domain. As a result, two-dimensional temperature fields and local heat transfer coefficients were obtained.
- The results obtained using the Fourier transform are consistent with those obtained with the Trefftz method, especially in the middle section of the minigap. Larger discrepancies observed near the inlet and outlet result from simplifying assumptions regarding the transform itself and the method used to determine the fluid temperature.
- The reconstructed wall temperature field showed very good agreement with the available measurements. The maximum mean relative difference was 0.78% for the smooth surface and 0.93% for the enhanced surface. For the fluid, validation was possible only at the inlet and outlet of the minigap; in these locations, the mean relative difference did not exceed 2% at the inlet and 1% at the outlet. Therefore, the reconstructed fluid temperature field should be regarded as consistent with the available experimental data, while its internal validation remains limited by the measurement layout.
- For both heating surface variants, similar axial distributions of the local heat transfer coefficient were observed over most of the length of the test section. The effect of surface enhancement was moderate under the present operating conditions. Slightly higher heat transfer coefficients for the threaded surface appeared only in selected axial regions and for selected heat flux values rather than as a pronounced overall enhancement over the entire minigap length. The results suggest that heat transfer in the analyzed system is governed primarily by flow boiling mechanisms rather than by surface microstructure effects.
- Uncertainty analysis showed that both the calculation method and the heater surface geometry influence the accuracy of HTC determination. The Fourier method provided the highest accuracy for the enhanced surface, while the Trefftz method was more stable for the smooth surface. Lower errors were obtained for saturated nucleate boiling, whereas subcooled boiling was characterized by greater scatter and higher errors, particularly at the channel inlet. Furthermore, the mean error decreased with increasing heat flux.
- The proposed methodology should be regarded as a reconstruction framework demonstrated for the present annular boiling dataset and operating window rather than as a universally validated predictive model. In particular, the friction factor fitting of Equation (4) remains preliminary because it was developed from a limited dataset.

A currently pursued research direction, extending earlier studies on flow boiling in annular minigaps with smooth and mechanically enhanced heated surfaces, including the configurations analyzed in [18], is the development of a miniaturized annular minichannel test section with heating surfaces modified both geometrically and chemically through coatings with controlled wettability [19,20]. The planned experimental program involves

surfaces with hydrophilic, hydrophobic, and superhydrophobic characteristics examined under otherwise comparable operating conditions. Such an approach is expected to provide a clearer distinction between the roles of surface topography and surface energy in boiling incipience, bubble nucleation, liquid rewetting near the wall, vapour removal, and the local thermal response of the flow. Because the characteristic dimensions are reduced, capillary forces, confinement effects, and surface–fluid interactions become more pronounced, making wettability an especially important parameter in the onset of nucleate boiling, wall superheat, bubble departure dynamics, flow stability, and local dryout. The new research direction therefore extends the existing studies beyond classical geometric enhancement and toward a broader surface engineering framework in which boiling performance is controlled jointly by microstructure and wettability.

Further work should also include a substantial expansion of the pressure drop dataset, a comparison with additional established two-phase pressure drop correlations, an extension of the operating range for both smooth and threaded surfaces, and additional temperature measurements within the test section to improve validation of the reconstructed fluid temperature field. This would make it possible to verify the robustness of Equation (4), assess its broader applicability, and better quantify the thermal and hydraulic effects of surface enhancement in annular minigap flow boiling. From an application-oriented perspective, future studies should also examine a wider range of surface modifications and operating conditions to determine when surface enhancement can yield a measurable and repeatable heat transfer benefit without an excessive pressure drop penalty.

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Conflicts of Interest: The authors declare no conflicts of interest.

Nomenclature

A	area of cross-section, m^2
A, B	constants
c_p	specific heat capacity, $\text{J}/(\text{kg K})$
D	constant
d_h	hydraulic diameter, m
$\mathcal{F}$	Fourier transform
f	Fanning friction factor
G	mass flux, $\text{kg}/(\text{m}^2\text{s})$
g	Trefftz functions
H	harmonic number
h	Trefftz function
HTC	heat transfer coefficient
L	length, m
$\mathcal{L}$	Laplace operator
MRD	mean relative difference

MRE	mean relative error
N	number of T-functions in approximation
p	pressure, Pa
q	heat flux, W/m ²
r	radius, m
Re	Reynolds number
RD	relative difference
T	temperature, K
u	Trefftz function
w	velocity, m/s
X_{th}	thermodynamic vapour quality
z	axial coordinate along the flow direction, m
Greek symbols	
α	heat transfer coefficient, W/(m ² K)
Δ	uncertainty, accuracy
λ	thermal conductivity, W/(m K)
m	dynamic viscosity, kg/(m s)
ρ	density, kg/m ³
ϕ^2	two-phase coefficient
Subscripts	
A	refers to acceleration pressure drop
ave	average
C	copper tube
F	refers to friction pressure drop
f	fluid
G	refers to gravity pressure drop
$meas$	measurement data
T	total flow resistance

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